

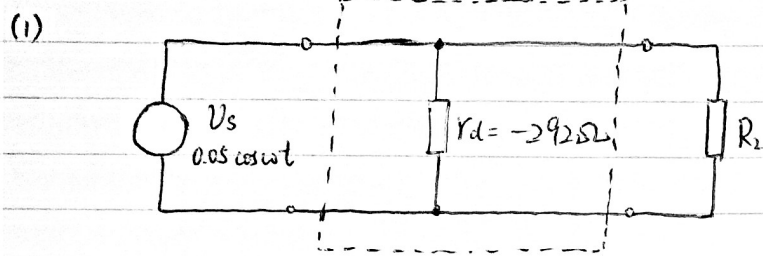
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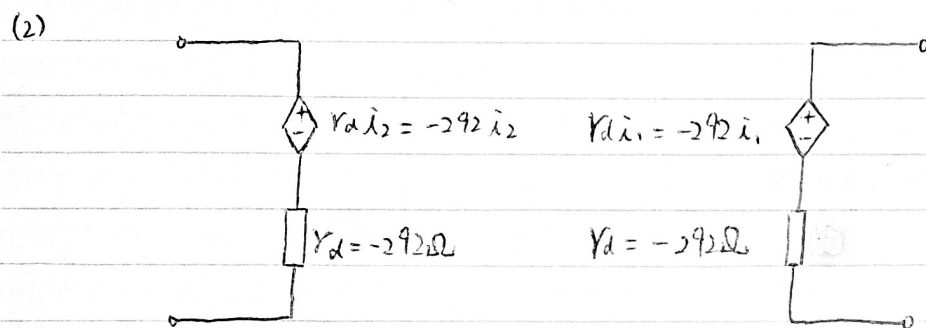
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作業 1



$$\mathbf{Z} = \begin{pmatrix} Y_d & Y_d \\ Y_d & Y_d \end{pmatrix} = \begin{pmatrix} -292 \Omega & -292 \Omega \\ -292 \Omega & -292 \Omega \end{pmatrix}$$

$$\mathbf{S} = \begin{pmatrix} \frac{R_{in} - R_s}{R_{in} + R_s} & 0 \\ 2 \sqrt{\frac{R_s}{R_L}} \frac{V_L}{V_s} & \frac{R_{out} - R_L}{R_{out} + R_L} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$



(3)

$$R_{in} = Z_{11} - \frac{Z_0 Z_{21}}{Z_{22} + R_L} = Y_d - \frac{Y_d^2}{Y_d + R_L} = \frac{Y_d R_L}{Y_d + R_L} = \frac{-292 \cdot 250}{-292 + 250} = 1.738 \text{ k}\Omega$$

$$R_{out} = Z_{22} - \frac{Z_0 Z_{21}}{Z_{11} + R_s} = Y_d - \frac{Y_d^2}{Y_d} = 0 \Omega$$

(4)

$$P_2 = \vec{\lambda}^T \vec{v} = \vec{\lambda}^T \mathbf{Z} \vec{\lambda} = Y_d (\hat{\lambda}_1^2 + 2 \hat{\lambda}_1 \hat{\lambda}_2 + \hat{\lambda}_2^2) = Y_d (\hat{\lambda}_1 + \hat{\lambda}_2)^2 < 0$$

該二端口網絡有源

作業 2

(1) 當 $\hat{\lambda}_3 = 0$ 時, $V_2 = -Z_0 \hat{\lambda}_1 = \hat{\lambda}_2 Y_d \Rightarrow V_1 = \hat{\lambda}_1 Z_0 = -\frac{Z_0^2}{Y_d} \hat{\lambda}_1 \Leftrightarrow Z_{11} = -\frac{Z_0^2}{Y_d}$

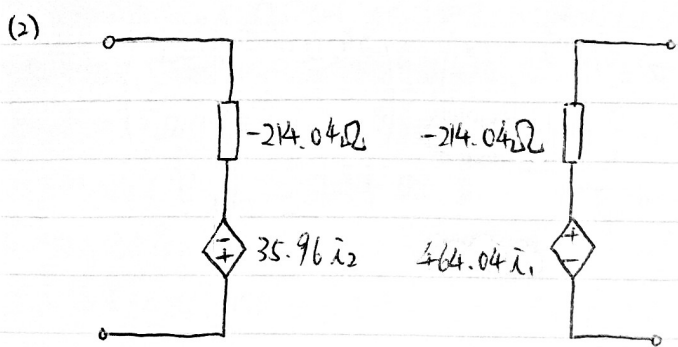
$$\Rightarrow V_3 = \hat{\lambda}_1 Z_0 - \hat{\lambda}_2 Z_0 = (Z_0 + \frac{Z_0^2}{Y_d}) \hat{\lambda}_1 \Leftrightarrow Z_{31} = Z_0 + \frac{Z_0^2}{Y_d}$$

當 $\hat{\lambda}_1 = 0$ 時, $V_2 = \hat{\lambda}_3 Z_0 = \hat{\lambda}_2 Y_d \Rightarrow V_1 = \hat{\lambda}_2 Z_0 - \hat{\lambda}_3 Z_0 = (\frac{Z_0^2}{Y_d} - Z_0) \hat{\lambda}_3 \Leftrightarrow Z_{12} = \frac{Z_0^2}{Y_d} - Z_0$

$$\Rightarrow V_3 = -\hat{\lambda}_2 Z_0 = -\frac{Z_0^2}{Y_d} \hat{\lambda}_3 \Leftrightarrow Z_{22} = -\frac{Z_0^2}{Y_d}$$

因此, $Z = \begin{pmatrix} -\frac{Z_0^2}{r_d} & \frac{Z_0^2}{r_d} - Z_0 \\ Z_0 + \frac{Z_0^2}{r_d} & -\frac{Z_0^2}{r_d} \end{pmatrix} = \frac{Z_0}{r_d} \begin{pmatrix} -Z_0 & Z_0 - r_d \\ r_d + Z_0 & -Z_0 \end{pmatrix} = \begin{pmatrix} -214.04\Omega & -35.96\Omega \\ 464.04\Omega & -214.04\Omega \end{pmatrix}$

$$S = \begin{pmatrix} 0 & \vec{I}^2 \\ \vec{I}^2 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 166.53 \\ 166.53 & 0 \end{pmatrix}$$



(3) $Z_{in} = Z_{11} - \frac{Z_{12}Z_{21}}{Z_{22} + Z_{21}} = -214.04 - \frac{464.04 \times (-35.96)}{250 - 214.04} = 250\Omega$

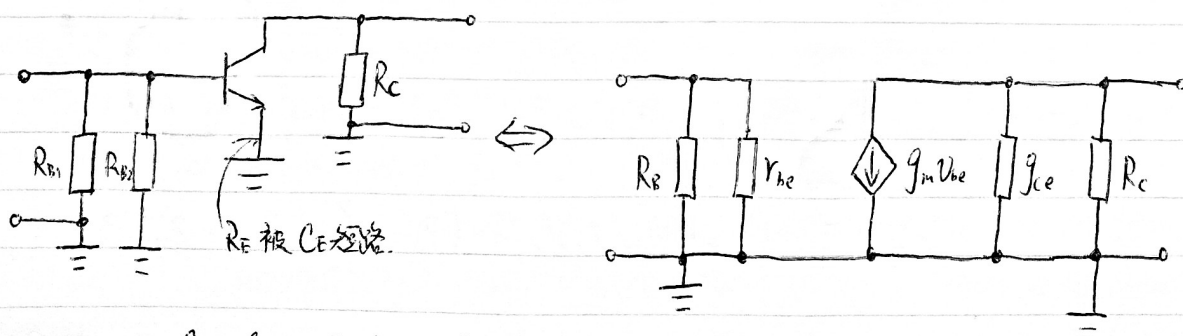
$$Z_{out} = Z_{22} - \frac{Z_{12}Z_{21}}{Z_{11} + Z_{12}} = 250\Omega$$

(4) $P_2 = \vec{i}^T Z \vec{i} = \frac{Z_0}{r_d} (-Z_0 i_1^2 + 2Z_0 i_1 i_2 - Z_0 i_2^2) = -\frac{Z_0^2}{r_d} (i_1 - i_2)^2 \leq 0$

該二端口網絡有源

作業3.

(1) $g_m = 41.5 \text{ mS}$, $g_{be} = \frac{g_m}{\beta} = 138.3 \mu\text{S}$, $g_{ce} = \frac{1.08 \text{ mA}}{700 \text{ V}} = 10.8 \mu\text{S}$



$$y_{11} = G_B + g_{be} = 256.2 \mu\text{S}, \quad y_{12} = 0$$

$$y_{21} = g_m = 41.5 \text{ mS}, \quad y_{22} = g_{ce} + g_c = 189.4 \mu\text{S}$$

$$Y = \begin{pmatrix} 256.2 \mu\text{S} & 0 \\ 41.5 \text{ mS} & 189.4 \mu\text{S} \end{pmatrix}$$

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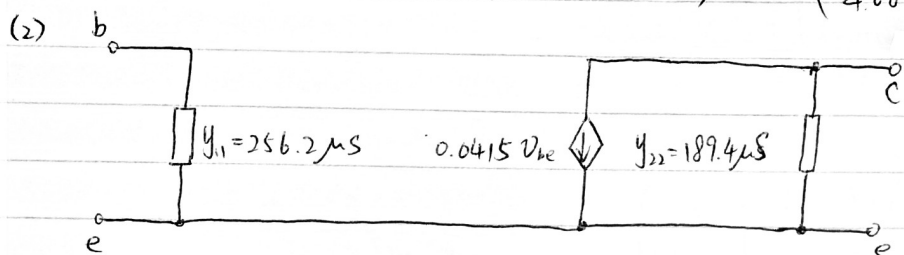
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$$S_{11} = \frac{R_{in} - R_s}{R_{in} + R_s} = 0.97$$

$$S_{22} = \frac{R_{out} - R_L}{R_{out} + R_L} = 0.98$$

$$S_{21} = 2 \sqrt{\frac{R_s}{R_L}} \frac{V_L}{V_s} = 2 \cdot \frac{V_L}{V_s} = 2 \times \frac{R_{in}}{R_{in} + R_s} \cdot 0.0415 \cdot \frac{1}{y_{11} + y_{12}} = 4.06$$

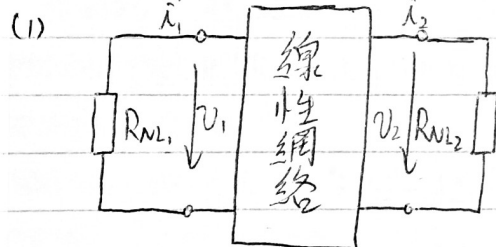
$$S_{12} = 0, \quad S = \begin{pmatrix} 0.97 & 0 \\ 4.06 & 0.98 \end{pmatrix}$$



(3) $R_{in} = \frac{1}{y_{11}} = 3.9 \text{ k}\Omega$, $R_{out} = \frac{1}{y_{22}} = 5.28 \text{ k}\Omega$

(4) 因為此 y 矩陣是正定矩陣，所以此二端口網絡無源。

作業 4



$$\begin{aligned} -\tilde{\lambda}_1 &= f_{NL1}(v_1) \\ -\tilde{\lambda}_2 &= f_{NL2}(v_2) \end{aligned} \quad \begin{pmatrix} \tilde{\lambda}_1 \\ \tilde{\lambda}_2 \end{pmatrix} = Y \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} + \begin{pmatrix} \tilde{\lambda}_{N1} \\ \tilde{\lambda}_{N2} \end{pmatrix}$$

$$\text{所以} \begin{pmatrix} f_{NL1}(v_1) \\ f_{NL2}(v_2) \end{pmatrix} + Y \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} + \begin{pmatrix} \tilde{\lambda}_{N1} \\ \tilde{\lambda}_{N2} \end{pmatrix} = 0$$

最適參量是 y

(2) $v_1 = f_{NL1}(-\tilde{\lambda}_1)$, $v_2 = f_{NL2}(-\tilde{\lambda}_2)$

$$\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = Z \begin{pmatrix} \tilde{\lambda}_1 \\ \tilde{\lambda}_2 \end{pmatrix} + \begin{pmatrix} v_{TH1} \\ v_{TH2} \end{pmatrix} \quad \text{所以} \begin{pmatrix} f_{NL1}(-\tilde{\lambda}_1) \\ f_{NL2}(-\tilde{\lambda}_2) \end{pmatrix} - Z \begin{pmatrix} \tilde{\lambda}_1 \\ \tilde{\lambda}_2 \end{pmatrix} - \begin{pmatrix} v_{TH1} \\ v_{TH2} \end{pmatrix} = 0$$

最適參量是 z

(3) $-\tilde{\lambda}_1 = f_{NL1}(v_1)$, $v_2 = f_{NL2}(-\tilde{\lambda}_2)$

$$\begin{pmatrix} \tilde{\lambda}_1 \\ v_2 \end{pmatrix} = g \begin{pmatrix} v_1 \\ \tilde{\lambda}_2 \end{pmatrix} + \begin{pmatrix} \tilde{\lambda}_{N1} \\ v_{TH2} \end{pmatrix} \quad \text{所以} \begin{pmatrix} f_{NL1}(v_1) \\ -f_{NL2}(-\tilde{\lambda}_2) \end{pmatrix} + g \begin{pmatrix} v_1 \\ \tilde{\lambda}_2 \end{pmatrix} + \begin{pmatrix} \tilde{\lambda}_{N1} \\ v_{TH2} \end{pmatrix} = 0$$

最適參量是 g

$$(4) \begin{pmatrix} f_{NL1}(-\lambda_1) \\ f_{NL2}(-\lambda_2) \end{pmatrix} = \begin{pmatrix} f_{NL1}(-I_0 - \Delta\lambda_1) \\ f_{NL2}(-I_{20} - \Delta\lambda_2) \end{pmatrix} = \begin{pmatrix} f_{NL1}(-I_0) + \frac{\partial f_{NL1}}{\partial \lambda_1}(-I_0) \cdot (-\Delta\lambda_1) + \dots \\ f_{NL2}(-I_{20}) + \frac{\partial f_{NL2}}{\partial \lambda_2}(-I_{20}) \cdot (-\Delta\lambda_2) + \dots \end{pmatrix}$$

$$\approx \begin{pmatrix} f_{NL1}(-I_0) \\ f_{NL2}(-I_{20}) \end{pmatrix} - \begin{pmatrix} \frac{\partial f_{NL1}(-I_0)}{\partial \lambda_1} & 0 \\ 0 & \frac{\partial f_{NL2}(-I_{20})}{\partial \lambda_2} \end{pmatrix} \begin{pmatrix} \Delta\lambda_1 \\ \Delta\lambda_2 \end{pmatrix}$$

直流分析時

$$\begin{pmatrix} f_{NL1}(-I_0) \\ f_{NL2}(-I_{20}) \end{pmatrix} - \begin{pmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{pmatrix} \begin{pmatrix} I_{10} \\ I_{20} \end{pmatrix} - \begin{pmatrix} V_{TH10} \\ V_{TH20} \end{pmatrix} = 0$$

不一致

交流分析時

$$\left[\begin{pmatrix} \frac{\partial f_{NL1}(-I_0)}{\partial \lambda_1} & 0 \\ 0 & \frac{\partial f_{NL2}(-I_{20})}{\partial \lambda_2} \end{pmatrix} + \begin{pmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{pmatrix} \right] \begin{pmatrix} \Delta\lambda_1 \\ \Delta\lambda_2 \end{pmatrix} + \begin{pmatrix} \Delta V_{TH1} \\ \Delta V_{TH2} \end{pmatrix} = 0$$

作業 5

$$(1) \frac{\partial i}{\partial v} = I_0 \cdot \frac{\partial}{\partial v} \left(\frac{\sinh \frac{v}{2V_T}}{\cosh \frac{v}{2V_T}} \right) = I_0 \frac{\left(\cosh^2 \frac{v}{2V_T} \right) \cdot \frac{1}{2V_T} - \left(\sinh^2 \frac{v}{2V_T} \right) \cdot \frac{1}{2V_T}}{\cosh^4 \frac{v}{2V_T}} = \frac{1}{2V_T} \cdot \frac{1}{\cosh^2 \frac{v}{2V_T}} \cdot I_0$$

$$= \frac{I_0}{2V_T} (1 - \tanh^2 \frac{v}{2V_T})$$

$$\frac{\partial^2 i}{\partial v^2} = -\frac{I_0}{2V_T} \cdot 2 \tanh \frac{v}{2V_T} \cdot \frac{1}{2V_T} \cdot (1 - \tanh^2 \frac{v}{2V_T})$$

因為 $|\tanh x| \leq 1$ ，所以最大直流工作點滿足 $\tanh \frac{v}{2V_T} = 0 \Leftrightarrow v = 0$

此時，微分電阻 $r_{d0} = \frac{\partial i}{\partial v}(0) = \frac{I_0}{2V_T}$

$$1\text{dB 線性範圍}: 20 \log \left(\frac{r_d}{r_{d0}} \right) = 1 \Leftrightarrow 20 \log \left(\cosh^2 \frac{v}{2V_T} \right) = 1 \Leftrightarrow \cosh \frac{v}{2V_T} = 10^{\frac{1}{40}}$$

$$e^{2x} - 2 \times 10^{\frac{1}{40}} e^x + 1 = 0 \Leftrightarrow (e^x - 10^{\frac{1}{40}})^2 = 10^{\frac{1}{20}} - 1 \Leftrightarrow e^x = 10^{\frac{1}{40}} \pm \sqrt{10^{\frac{1}{20}} - 1}$$

$$v = 2V_T \ln \left(10^{\frac{1}{40}} \pm \sqrt{10^{\frac{1}{20}} - 1} \right) = 17.81\text{mV} \text{ 或 } -17.81\text{mV}$$

所以 1dB 工作範圍是 $-17.81 \sim 17.81\text{mV}$

$$(2) \frac{dy}{dx} = K_a \cos x, \quad \frac{d^2 y}{dx^2} = -K_a \sin x, \quad \text{直流工作點位置是 } x = k\pi (k \in \mathbb{Z})$$

此時的斜率是 $\frac{dy}{dx}(k\pi) = K_a \cos(k\pi)$, 1dB 工作範圍是 $20 \log |\cos x| \leq -1$

$$\cos x = \pm 10^{-\frac{1}{20}}, \quad x = \pm 27^\circ + 180^\circ k$$

其直流工作範圍是 $-27^\circ + 180^\circ k \sim 27^\circ + 180^\circ k (k \in \mathbb{Z})$

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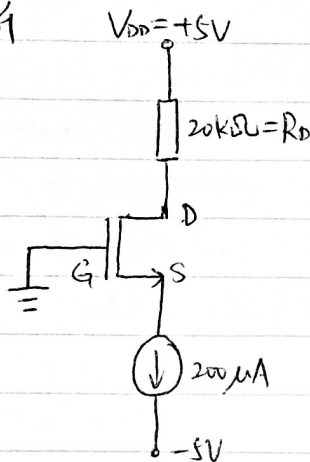
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習題 6

直流分析



$$V_{D0} = V_{DD} - I_{D0} R_D = 5V - 200\mu A \cdot 20k\Omega = 1V$$

$$I_{D0} = \beta (V_{GS0} - V_{TH})^2 \left(1 + \frac{V_{D0} + V_{GS0}}{V_E}\right)$$

$$0.2mA = 1mA/V^2 \cdot (V_{GS0} - 1V)^2 (1.05 + 0.05 V_{GS0})$$

$$\Leftrightarrow 0.2 = (V_{GS0} - 1)^2 (1.05 + 0.05 V_{GS0})$$

$$= V_{GS0}^2 - (1.1 + 0.05 V_{GS0})$$

$$\Leftrightarrow 0.05 V_{GS0}^3 + 1.1 V_{GS0}^2 - 0.2 = 0$$

$$\Leftrightarrow V_{GS0}^3 + 22 V_{GS0}^2 - 4 = 0$$

$$V_{GS0}^{(1)} = 0.43V$$

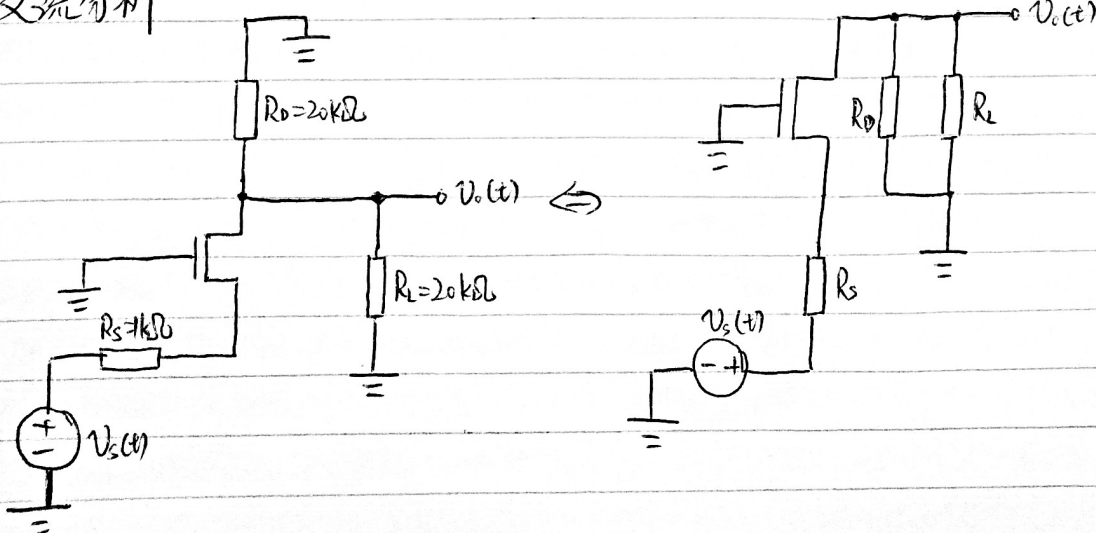
$$\text{記 } f(v) = v^3 + 22v^2 - 4, \quad V_{GS0}^{(2)} = V_{GS0}^{(1)} - \frac{f(V_{GS0}^{(1)})}{f'(V_{GS0}^{(1)})} = 0.422$$

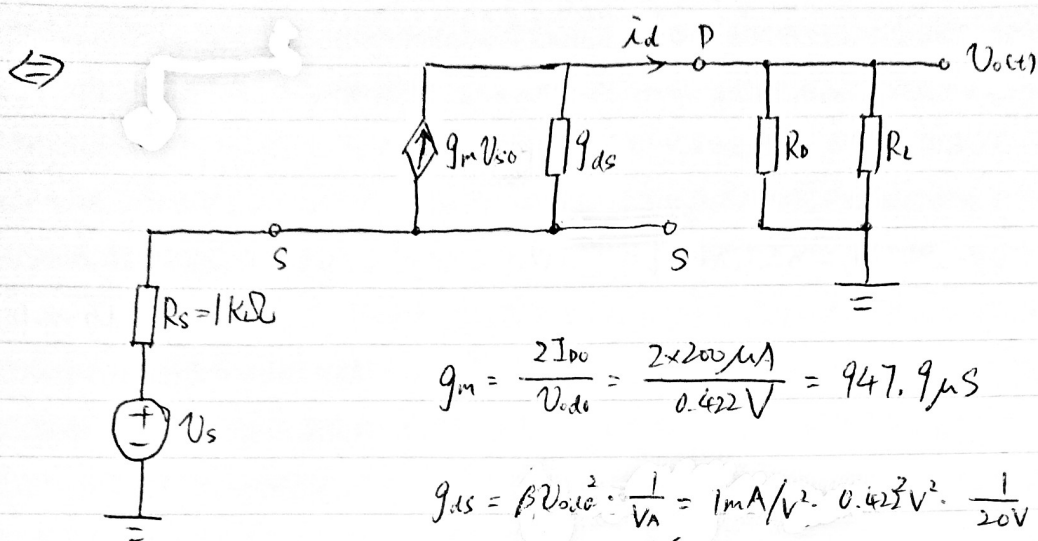
$$V_{GS0}^{(3)} = V_{GS0}^{(2)} - \frac{f(V_{GS0}^{(2)})}{f'(V_{GS0}^{(2)})} = 0.422 > 0.$$

$$\Leftrightarrow V_{GS0} = 1.422V$$

又知 $V_{GS0} = -1V < V_{TH0} = 1V$, 所以此 NMOS 工作在恆流區

交流分析





$$g_m = \frac{2I_{DQ}}{V_{ov}} = \frac{2 \times 200 \mu A}{0.422 V} = 947.9 \mu S$$

$$g_{ds} = \beta V_{ov}^2 \cdot \frac{1}{V_A} = 1 mA/V^2 \cdot 0.422^2 V^2 \cdot \frac{1}{20 V} = 8.9 \mu S$$

$$V_o = i_d (R_D // R_L), \quad (V_{gs} - V_o) g_{ds} + g_m V_{gs} = i_d$$

$$V_{gs} = V_s - i_d R_s$$

$$(V_s - i_d R_s) (g_{ds} + g_m) - V_o g_{ds} = i_d$$

$$(V_s - \frac{V_o}{R_D // R_L} R_s) (g_{ds} + g_m) - V_o g_{ds} = \frac{V_o}{R_D // R_L}$$

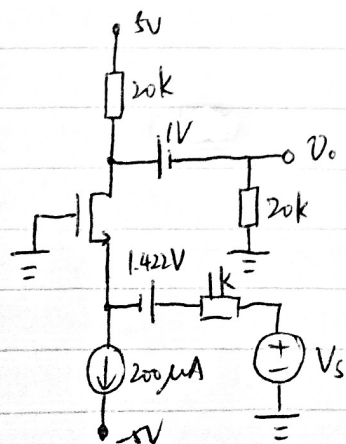
$$V_s (g_{ds} + g_m) = V_o \left(\frac{1}{R_D // R_L} + \frac{R_s (g_{ds} + g_m)}{R_D // R_L} + g_{ds} \right)$$

$$A_v = \frac{V_o}{V_s} = \left(\frac{g_D + g_L}{g_{ds} + g_m} + \frac{R_s}{R_D // R_L} + \frac{g_{ds}}{g_{ds} + g_m} \right)^{-1} = 4.68$$

$$G_p = \frac{\frac{V_o^2}{R_L}}{\frac{V_s^2}{4R_s}} = 4 \frac{R_s}{R_L} \cdot A_v^2 = 4.37$$

選作:

(1) 無交流小信號時, NMOS 吸收的功率 $P_{DQ} = V_{DQ} I_{DQ} = 2.422 V \times 200 \mu A = 0.4844 mW = 484.4 \mu W$



有交流小信號時

$$NMOS 吸收的功率, P_{DQ} = (2.422 - V_s) (200 \mu A - i_d) < P_{DQ0}$$